Lecture 1

Basic Concepts and Set Theory

09.02.2026

Outline

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Sample space

Definition

A sample space S is the set of all possible outcomes of an random experiment.

Examples:

- 3 coin tosses
- One dice roll
- Three dice roll
- Sum of two rolls
- Coin toss until first Head
- Lifetime of a transistor

Examples (Cont)

- Measurement of the voltage output v from a transducer, the maximum and minimum of which are $+5$ and -5 volts.

- Pick (x, y) from inside the unit circle centered at the origin.

- Pick a point from inside the triangle defined by the vertices $(0, 0)$, $(0, 1)$, and $(1, 0)$.

Exercise (Sample Space)

Select a point in the interior region bounded by the lines $x + y = 1$, $x - y = 1$, $-x + y = 1$, and $-x - y = 1$.

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	Even number	E $= \{$2,4,6$\}$
	$<$ 2 minutes	E $= [0, 120)$

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- Impossible event: empty set $\emptyset \subset S$
- Certain event: $S \subset S$

Outline

Set theory

Let A, B be two events.

Definition

1. **Intersection** $A \cap B$: *implies the event that both A and B occur*
2. **Union** $A \cup B$: *implies the event that at least one of A or B occur*
3. *The* **complement** *of an event A denoted A^c (also notated A' or $\bar{A}$):* $A^c = S \setminus A$ *- the event that A does not occur*
4. $A \subset B$ *implies that the occurrence of A implies the occurrence of B*
5. $A = B$ *if and only if $A \subset B$ and $B \subset A$.*

Venn diagram

More set theory

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Two events A and B are **mutually exclusive** *or* **disjoint** *if they have no outcomes in common, i.e. $A \cap B = \emptyset$.*

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Examples of exclusive events?

Some rules

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3. Distributive laws

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DeMorgan's Laws

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  Left $\subset$ Right $\Longleftrightarrow$ For any $x \in (A \cup B)^c$, then $x \in A^c \cap B^c$.
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- DeMorgan's laws can be generalized to n events $A_1, \ldots, A_n$:

$$\left(\bigcup_{i=1}^{n} A_i\right)^c = \bigcap_{i=1}^{n} A_i^c, \quad \left(\bigcap_{i=1}^{n} A_i\right)^c = \bigcup_{i=1}^{n} A_i^c$$

- Suppose $A_i \subset S, \quad i = 1, 2, \ldots.$ Then

$$\bigcup_{i=1}^{\infty} A_n = \{x \in S | x \in A_i \text{ for some } i = 1, 2, \ldots, \}$$

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Examples:

- Simplify $\bigcup_{i=1}^{\infty}[0, i]$
- Simplify $\bigcap_{i=1}^{\infty}[-1/i, 1/i]$

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- Partitions are useful for chopping up sets into manageable, disjoint pieces. Given a set A, write

$$\begin{aligned} A &= A \cap S \\ &= A \cap \left(\bigcup_{i=1}^{n} B_i \right) = \bigcup_{i=1}^{n} (A \cap B_i) \end{aligned}$$

Exercise

Sketch the following subsets of the x-y plane

- $A_z = \{(x, y) | \max(x, y) \leq z\}$ for $z = 3$
- $B_z = \{(x, y) | \min(x, y) \leq z\}$ for $z = 3$
- $A_2 \cap B_3$
- $A_4 \cap B_3$